ETH zürich

ALMA MATER STUDIORUM
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CachePool: Many-core cluster of customizable, lightweight scalar-vector PEs for irregular L2 data-plane workloads

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PULP Platform
Open Source Hardware, the way it should be!

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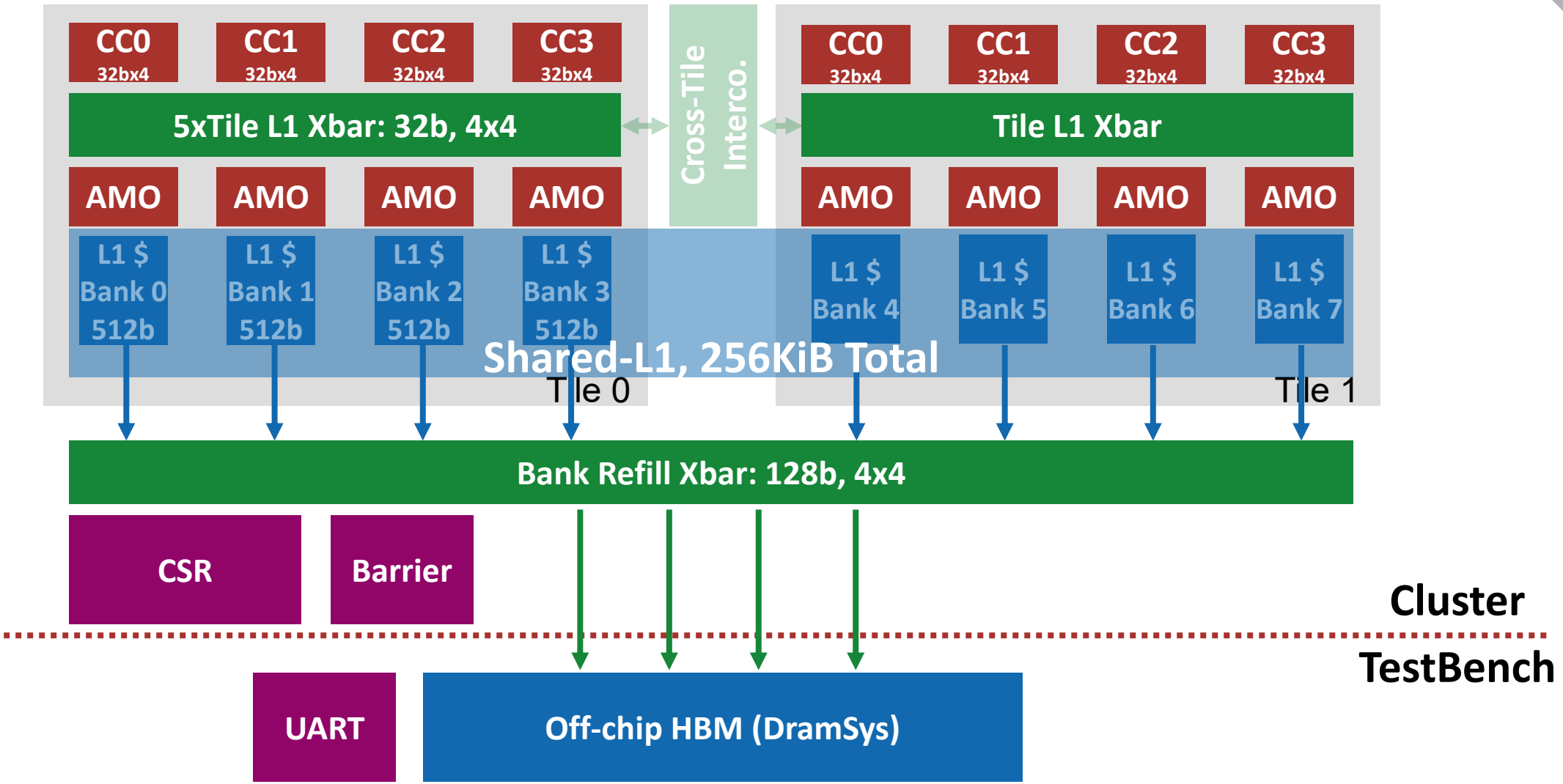
Hardware Development (Cluster)



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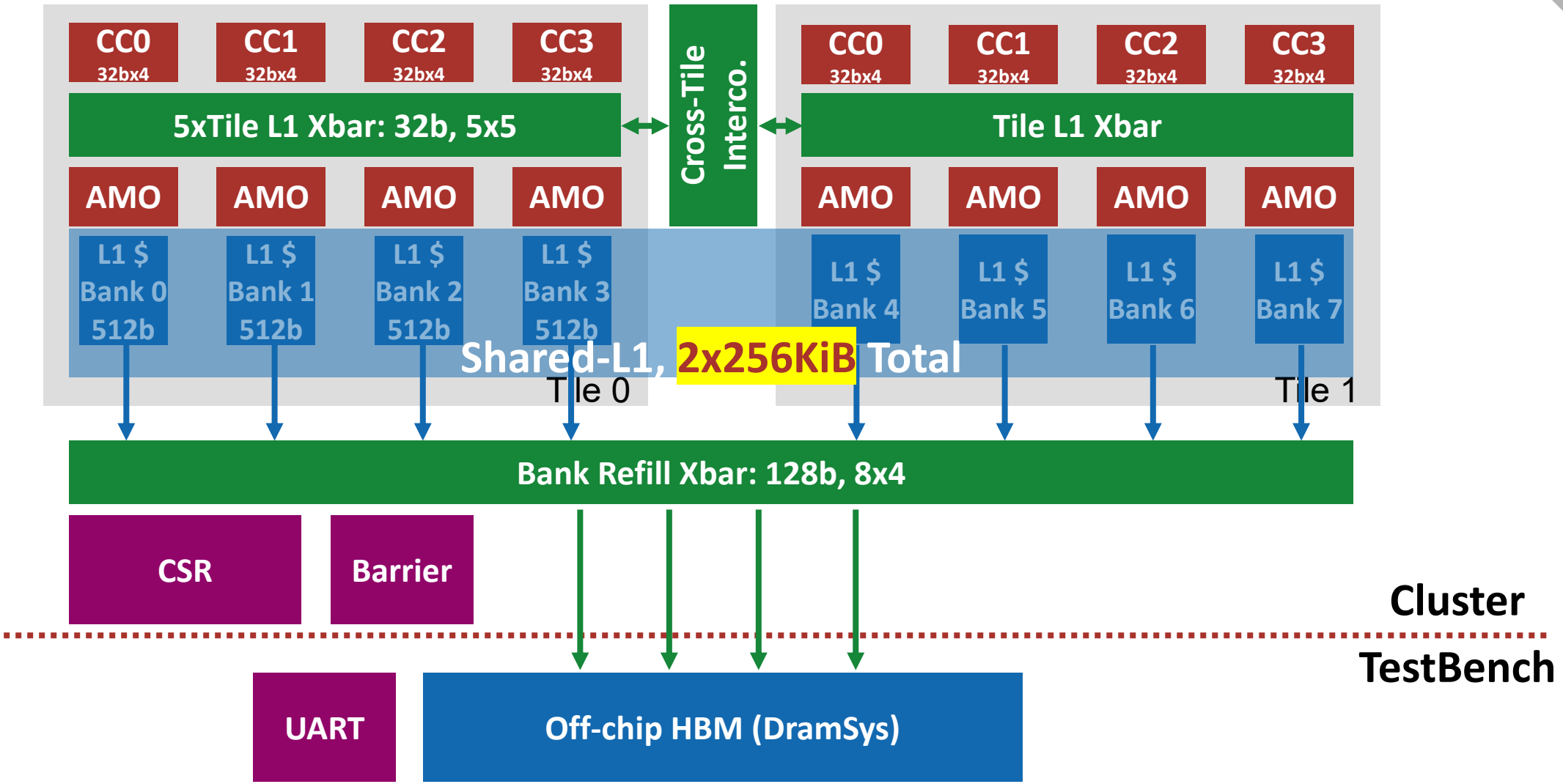

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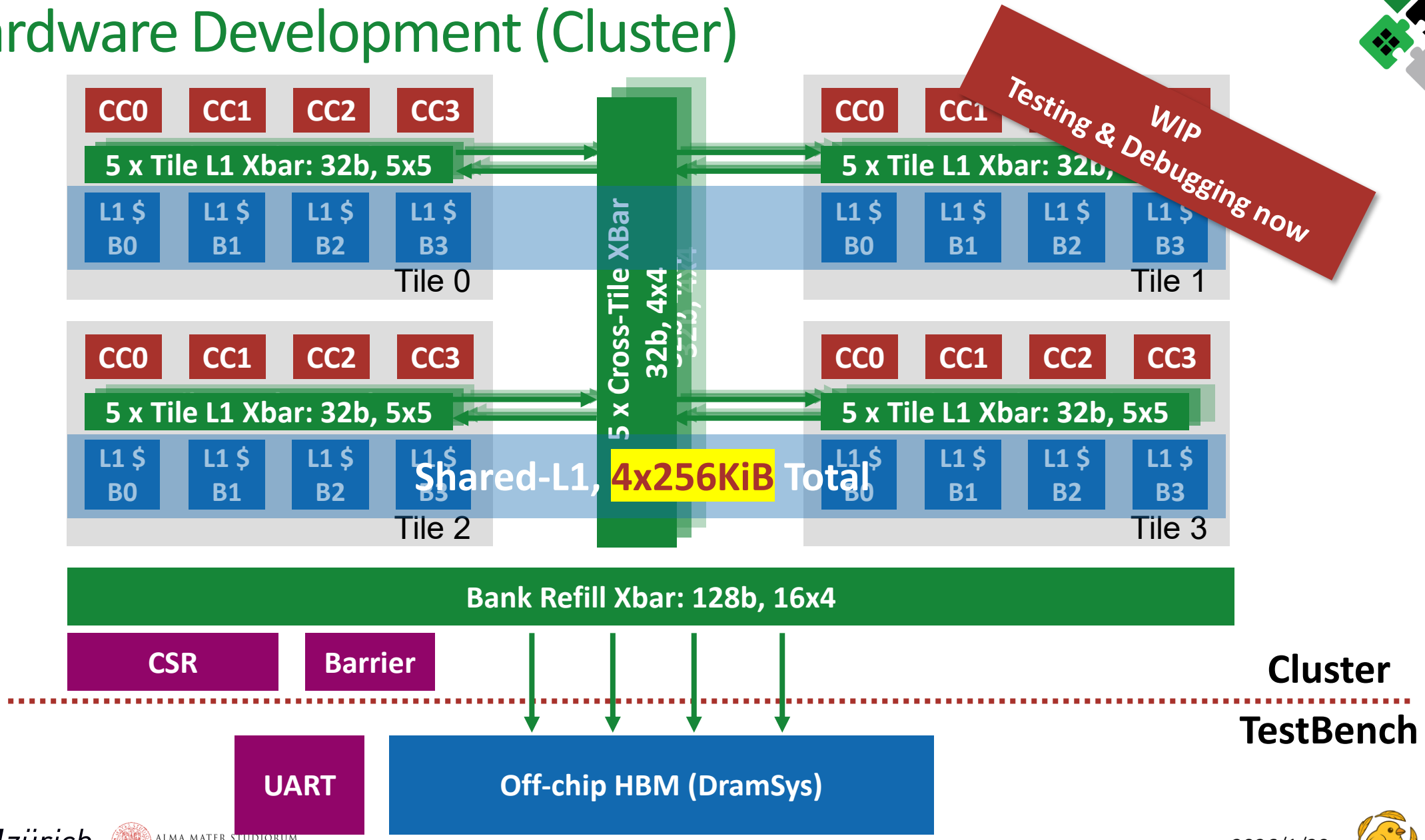
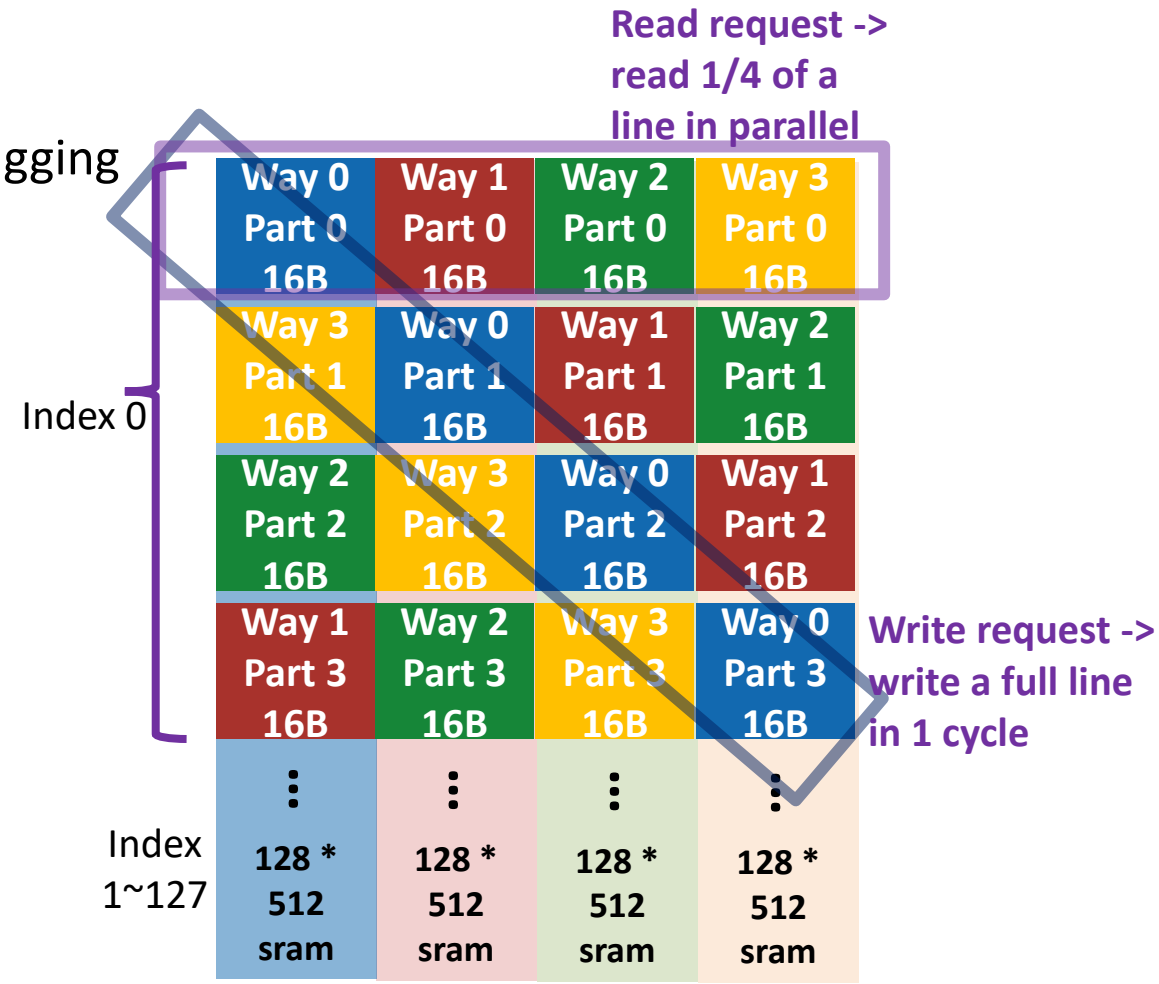


Hardware Development (Cluster)

- **WIP: Multi-Tile Configuration**
  - Two-tile configurations are fully working now
    - Cross-tile interconnection implemented
    - Caches are correctly shared and banked
    - Micro-kernels can execute correctly
  - Now working on X-Tile (power of 2) configurations
    - Some crossbar optimization is needed, e.g. change the 16x4 bank refill xbar into 2-level xbars inside the tile
    - Benchmarking: tune the kernels to make use of the available bandwidth
    - Backend verification

Hardware Development (L1 Cache)

- **Initial implementation of folded cache**
  - Simple tests have passed
  - Failed with complex accessing patterns, now debugging



Next Step

- **Diyou Side**
  - Finish the multi-tile part
    - RTL
    - Benchmarking
    - Backend
  - Start cache partitioning work
- **Zexin Side**
  - Finish folded cache design
  - Start the GVSoC modeling

Timeline

		2025			2026								
		11	12	1	2	3	4	5	6	7	8	9	10	11
System	Multi-Tile Scaling [RTL]													
	4/16-Group Scaling [GVSoC + RTL]													
L1 Access Latency	Hardware optimization (Cache & Interco) [RTL]													
	Cache partitioning [RTL]													
	WT-L1 by master student [RTL]													
Large NoC latency	Reduce per-hop latency [Physical]													
	Cut clk-domain (optional) [RTL + Physical]													
	NoC Topologies [RTL + GVSoC + Physical]													

Timeline

		2025			2026								
		11	12	1	2	3	4	5	6	7	8	9	10	11
Improve scalar core performance	Shared vector core [RTL + GVSoC]													
SW Kernel	Kernel optimization [SW]													
	Kernel development [SW]													
Report	Final Report and Benchmark													

L1 Access Latency

- **Problem: increased L1 access latency when scaling up**
- **Solutions**
  - **(Ongoing, RTL & Physical) Hardware optimization**
    Further optimize the interconnection and cache controller to reduce access latency on existing architecture
    => Retiming the current interconnection, remove unnecessary cuts in interco and cache
    => Target to reduce 1-2 cycles (~5 cyc) for hit-to-use latency
  - **(Ongoing, RTL) Cache partitioning**
    Partition the shared L1 to keep data localized to avoid remote access latency
  - **(Ongoing, RTL & Physical) WT-L1**
    Explore and evaluate a private WT L1 for each core
    => Ongoing master thesis, now is working on the coherence between L1 caches
    => Currently plan to use hierarchical directory-based approach

L1 Access Latency

- **Problem: large NoC latency (~28 cyc based on previous exploration)**
- **Solutions**
  - **(Ongoing, Physical) Reduce hop latency**
    Explore the timing under 7nm technology to see the possibility of reducing per-hop latency to 1 cycle
    => Reduce the latency by 1/2
  - **(Planned, Physical) Cut clk-domain (optional)**
    If 1-cyc is not possible, explore to operate the NoC under a higher frequency than cores (e.g. 1.5GHz)
    => Effectively cut the latency by 1/3
  - **(Planned, GVSoC & Physical) NoC Topology**
    Explore different NoC topologies
    => Torus to reduce the diameter of NoC by 1/2, reduce longest latency by 1/2

PE-Related

- **Problem: Improve scalar core performance**
- **Solutions**
  - **(Planned, RTL & GVSoC) Shared vector core**
    Explore two Snitch sharing one Spatz configuration to improve scalar performance
    Software complexity needs to be considered, especially on the possible conflicting use of register file.

RLC Workload

- **Problem: Current RLC model cannot reveal all properties of the kernel**
- **Solutions**
  - **(Ongoing, SW) Kernel optimization**
    Adapt the current model to better represent the real cases, e.g. do some operations on the data. Further cooperation with HQ side to develop a more complete and better model.
  - **(Planned, SW) Kernel development**
    Implement the missing use cases for multi-user

Thank you!